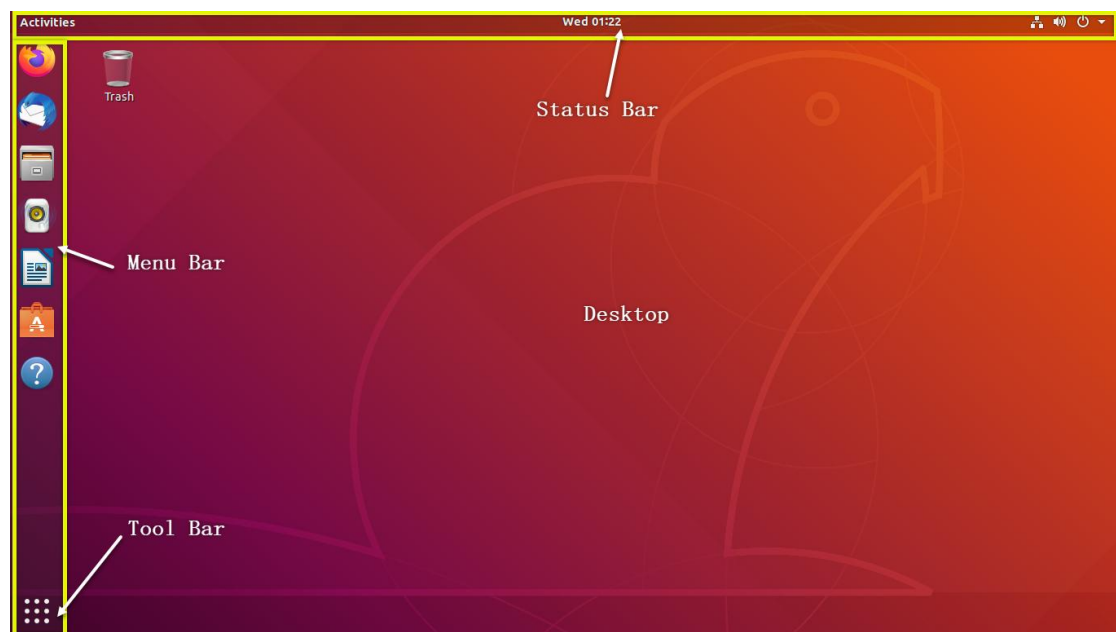


Lesson 4 Basic Operation of Linux

1.GUI Layout

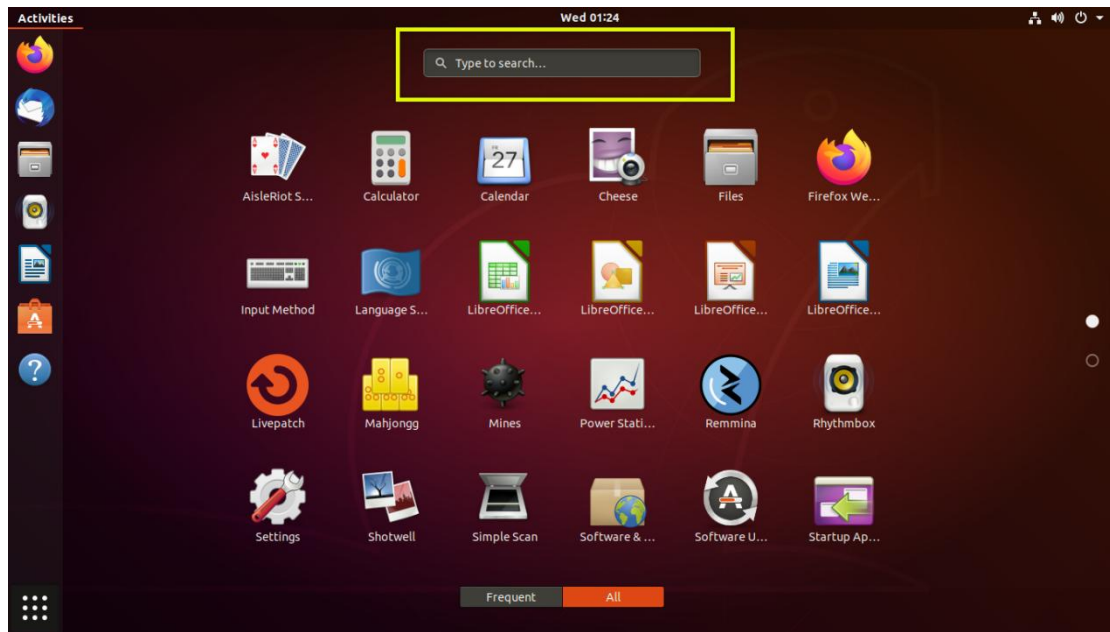
Take GUI of Ubuntu for example. If you want to check GUI of other Linux versions, please search online.

The yellow frame at left is the menu bar, at right is the desktop, and under it is the tool bar. And the status bar is on the top.

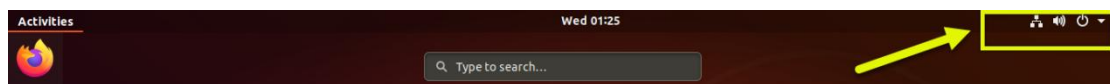


Some basic tools including, browser, mailbox and Office software, can be found in the menu bar, which can meet your basic usage needs.

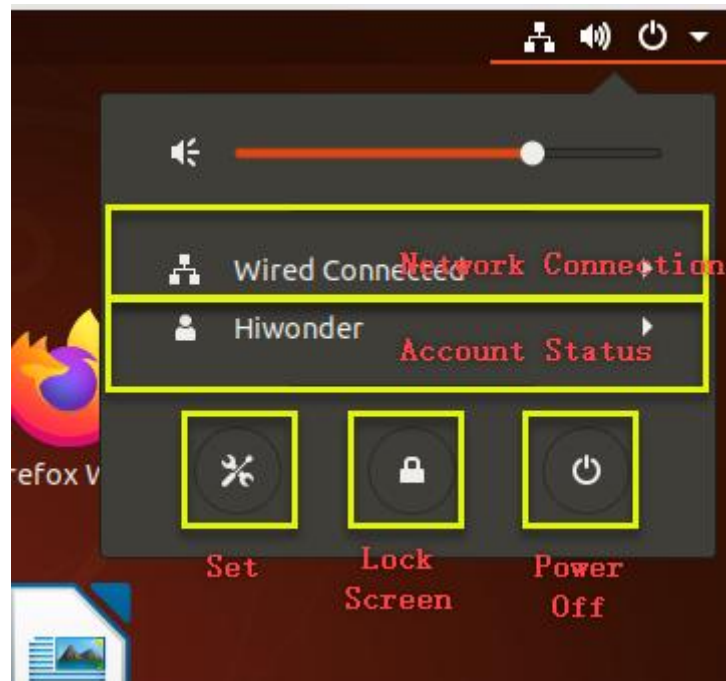
1) Click “”, and we can select the various kinds of tools. You can input the tool name in the input field to find the corresponding tool.



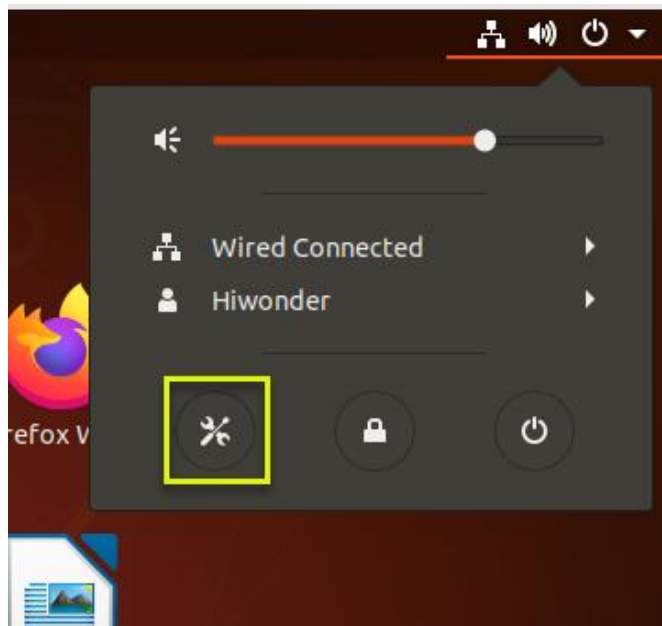
2) Click the far right corner of the status bar.



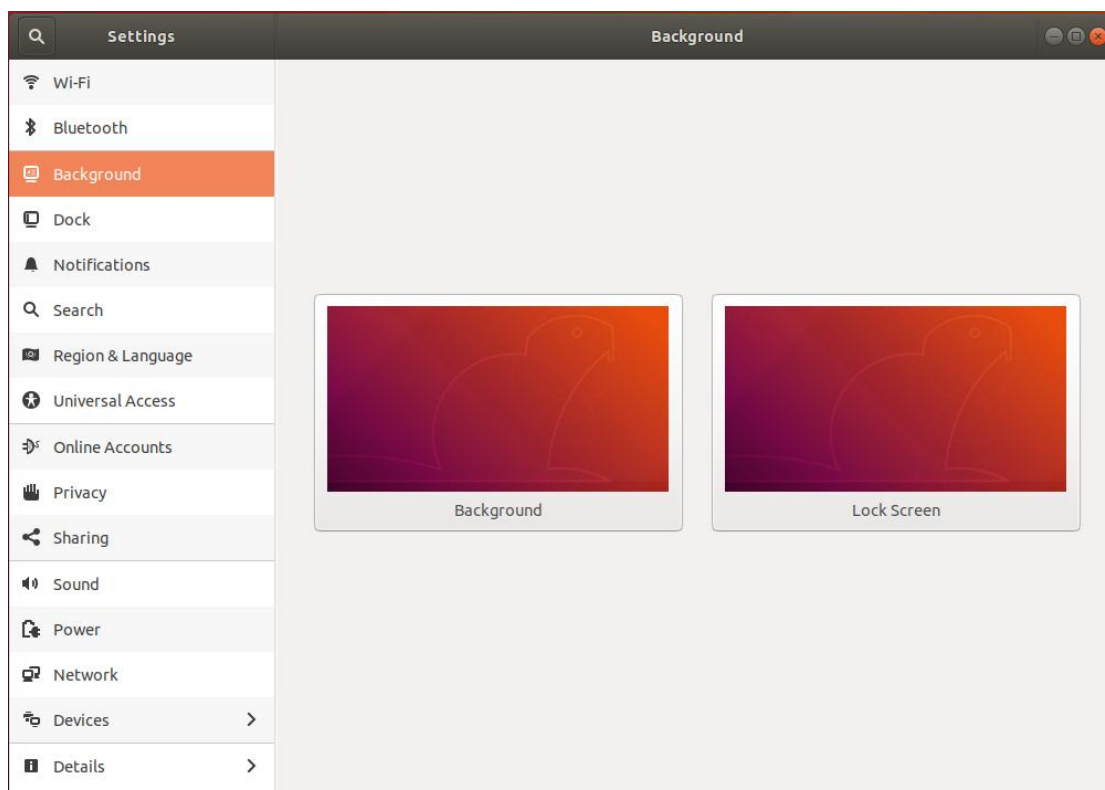
3) The window below will appear.



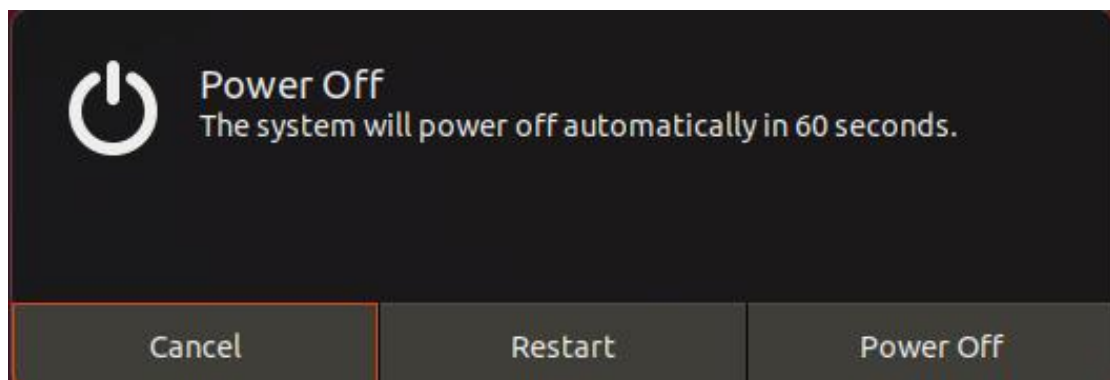
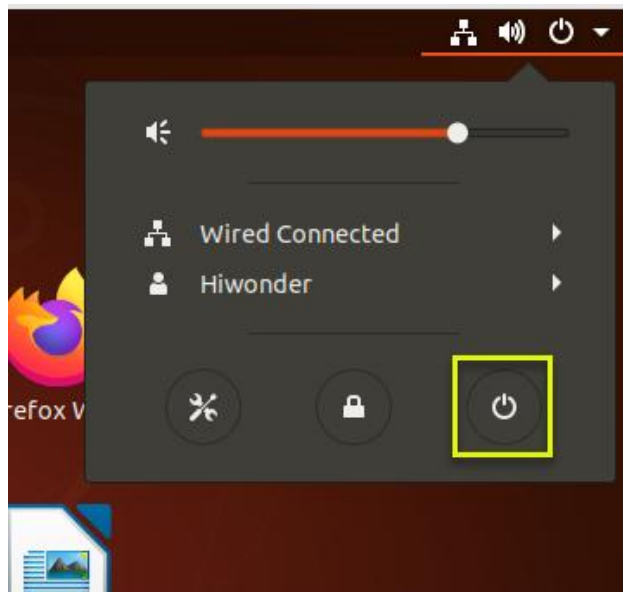
4) Click “”



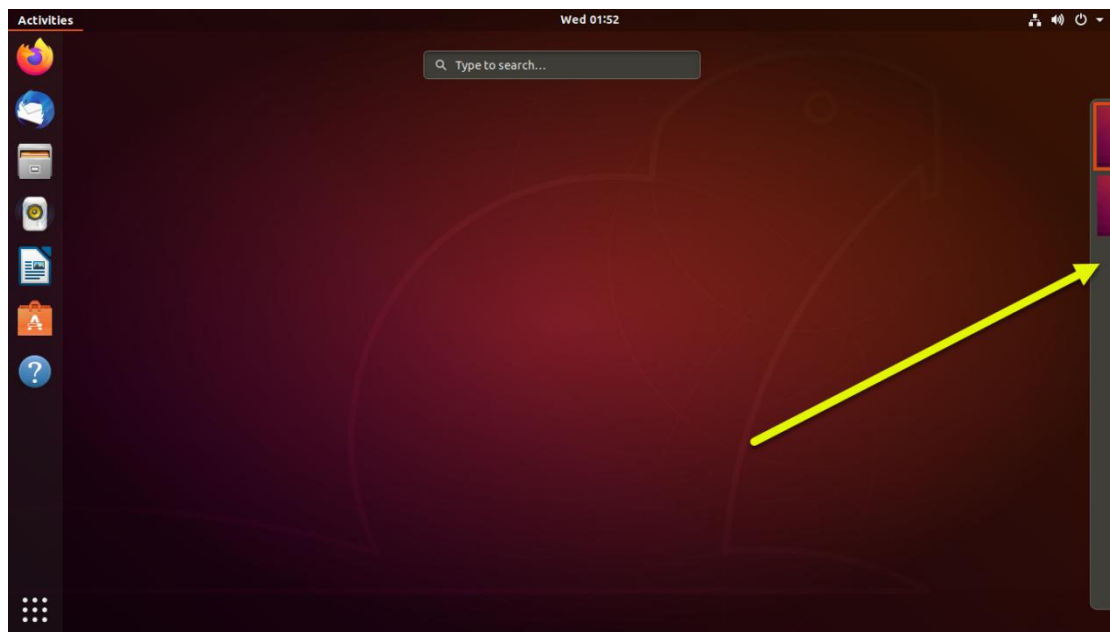
5) You can set Ubuntu system in this interface.



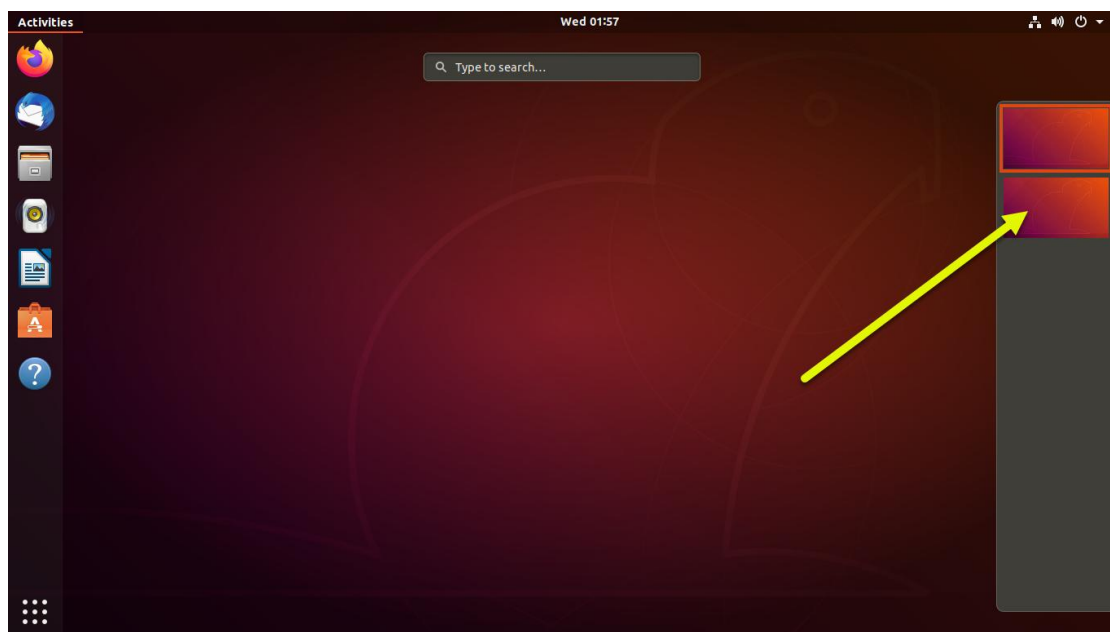
6) Click , and then you can shutdown or restart the system.



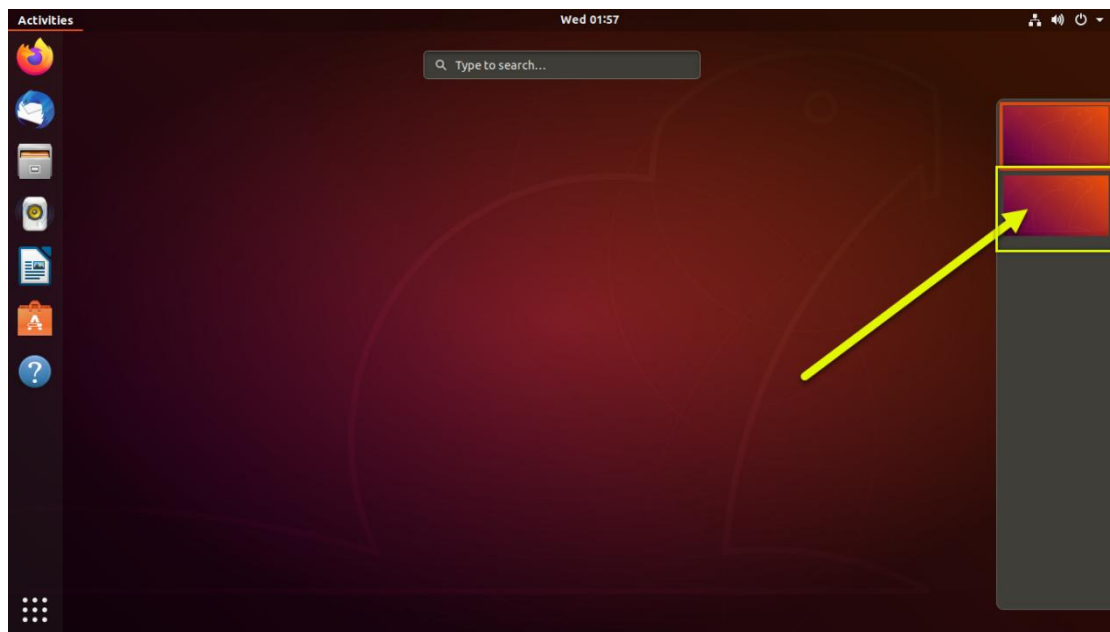
7) Press “win” key, and move the cursor to the far right.



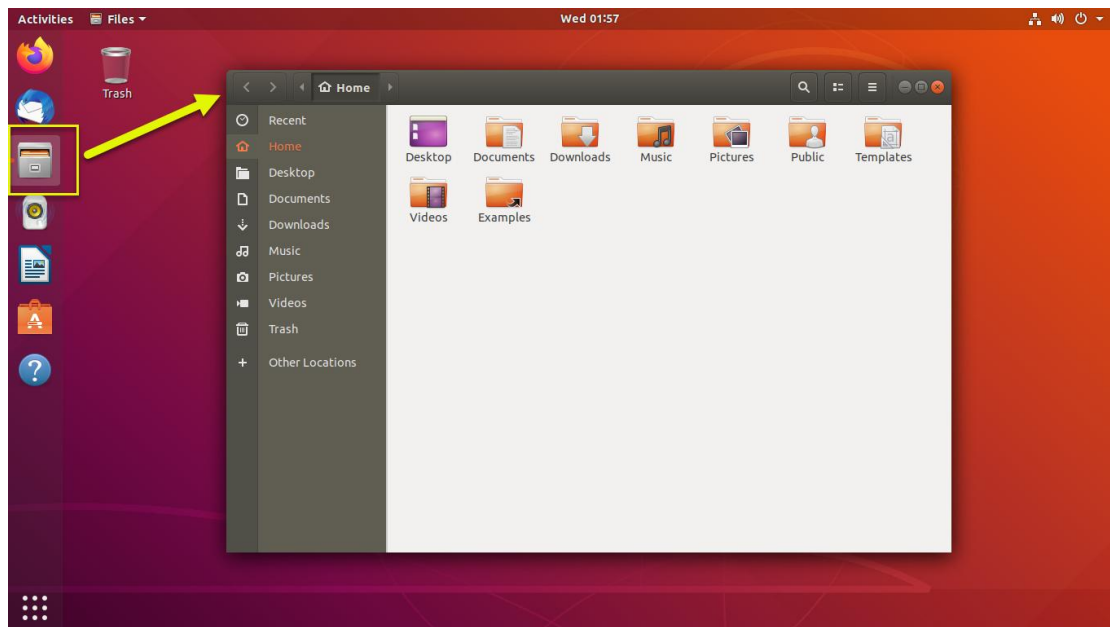
8) Click the second desktop framed in yellow.



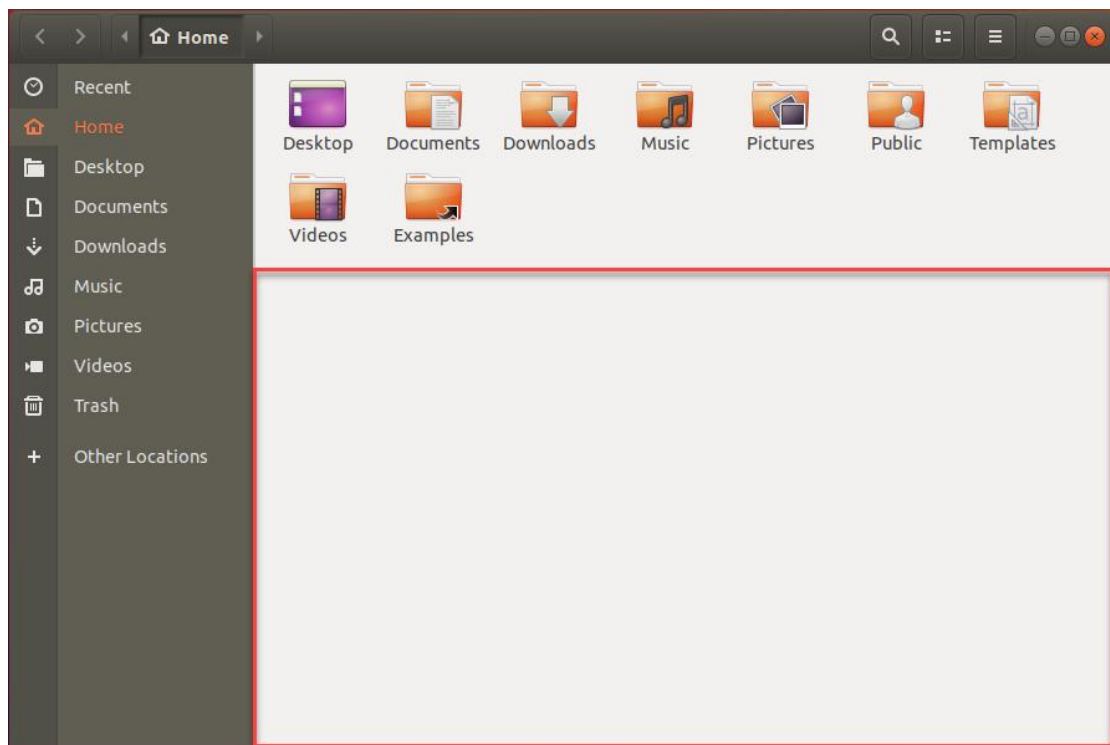
9) Then you can switch it to a new desktop.



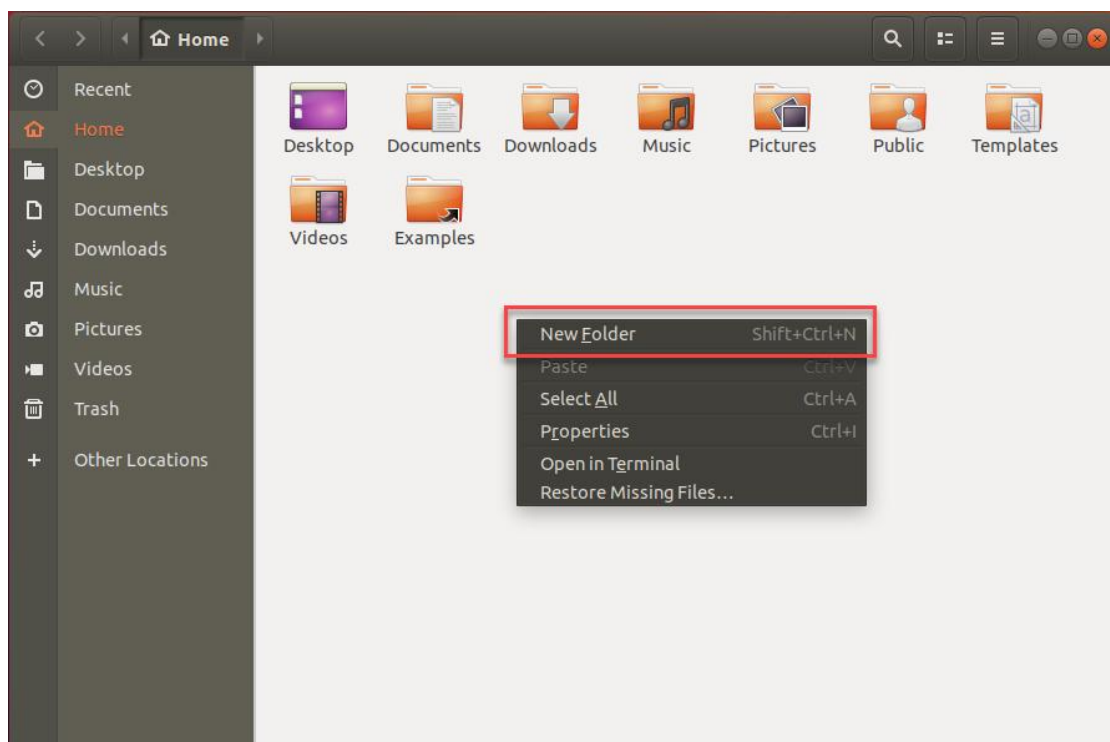
10) Click the folder icon at left to open file manager, as the picture below shown.



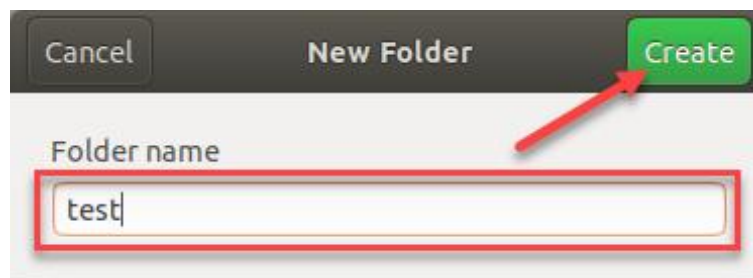
11) Right click the blank area in red frame.



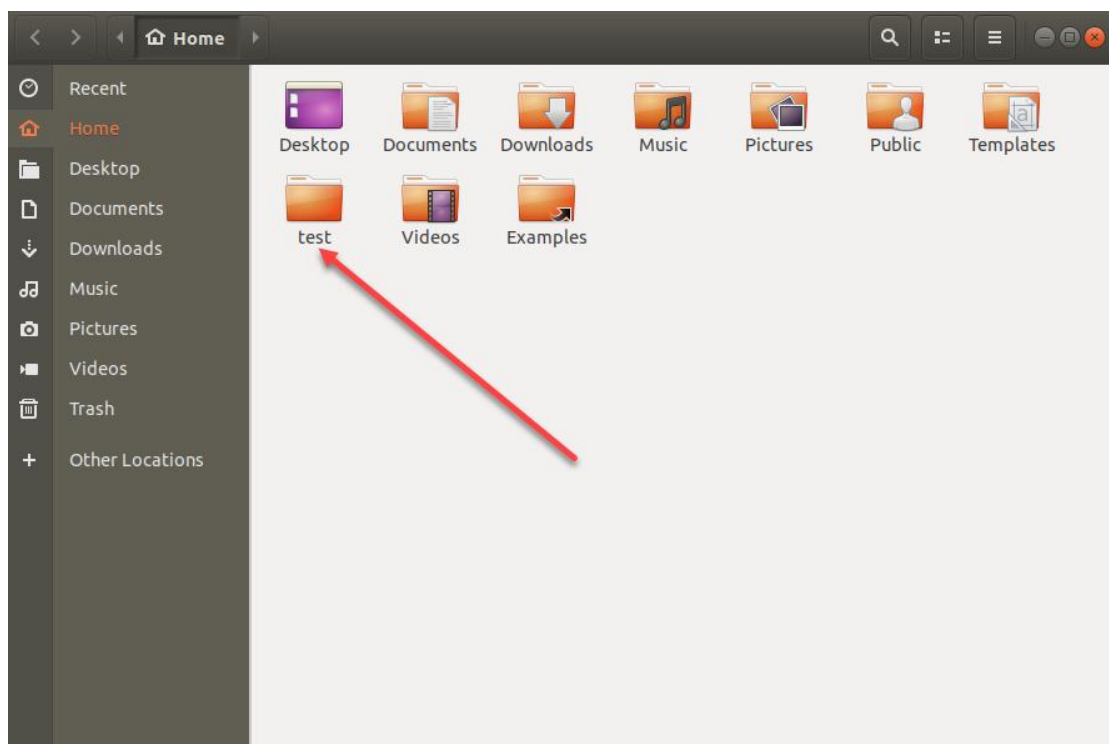
12) Click “New Folder”.



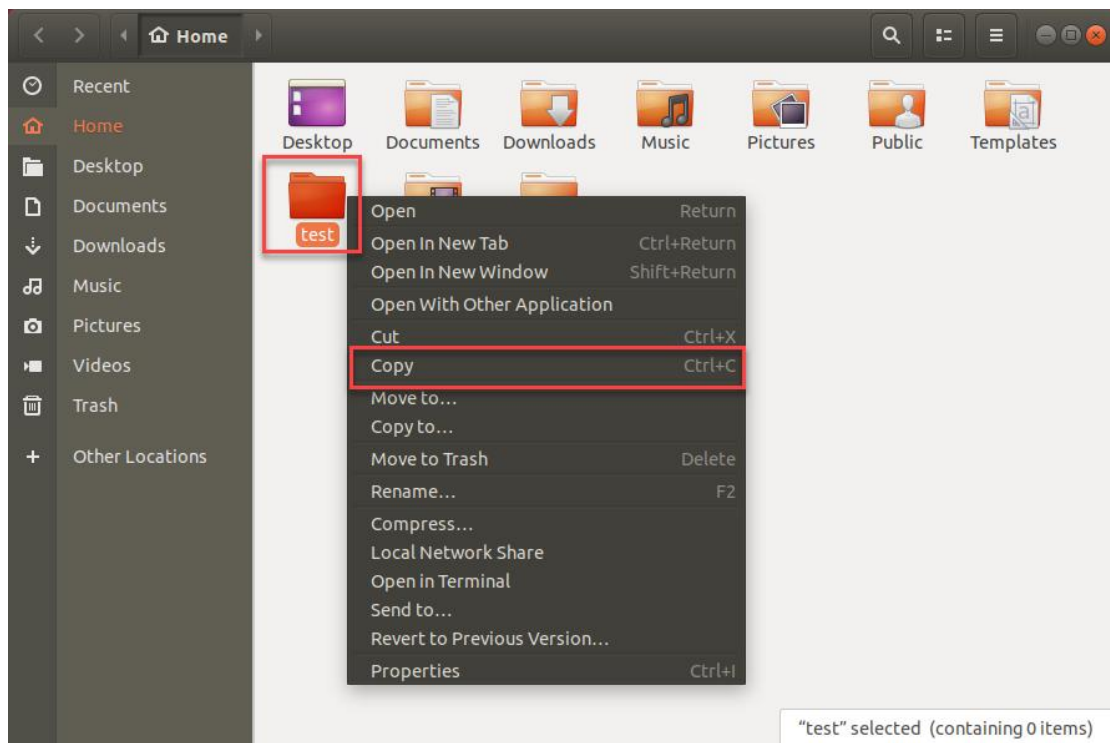
13) Input the folder name, for example “test”. After that, click “add”.



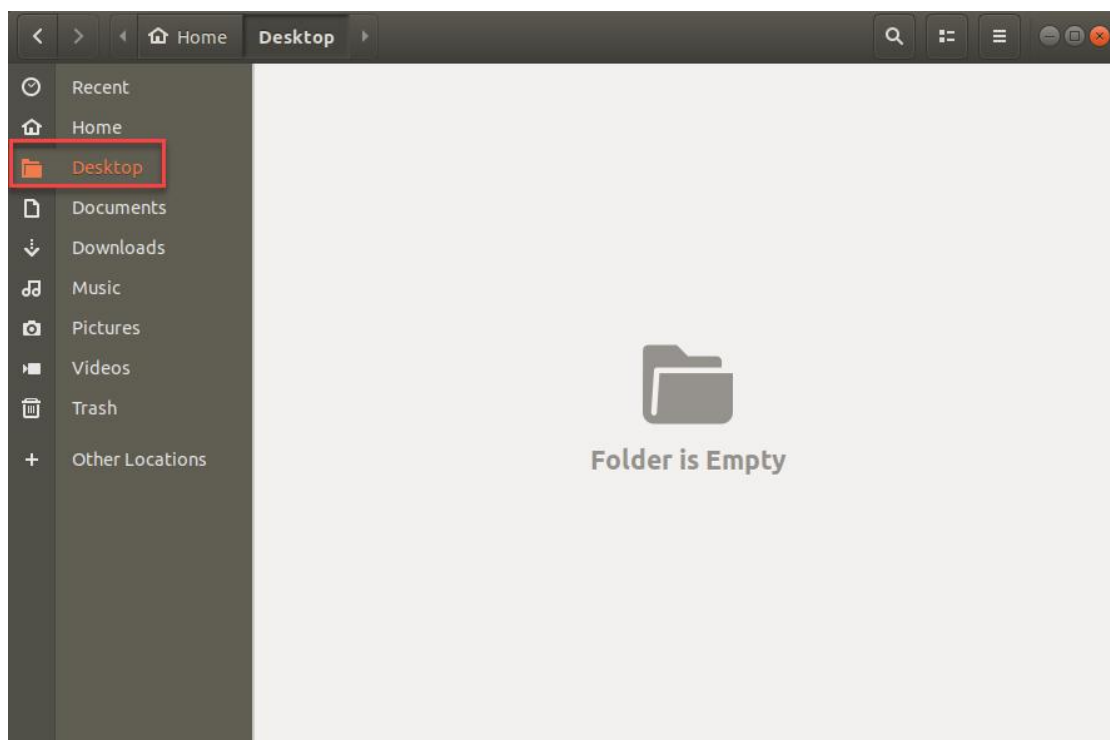
14) The new folder is as the picture shown.



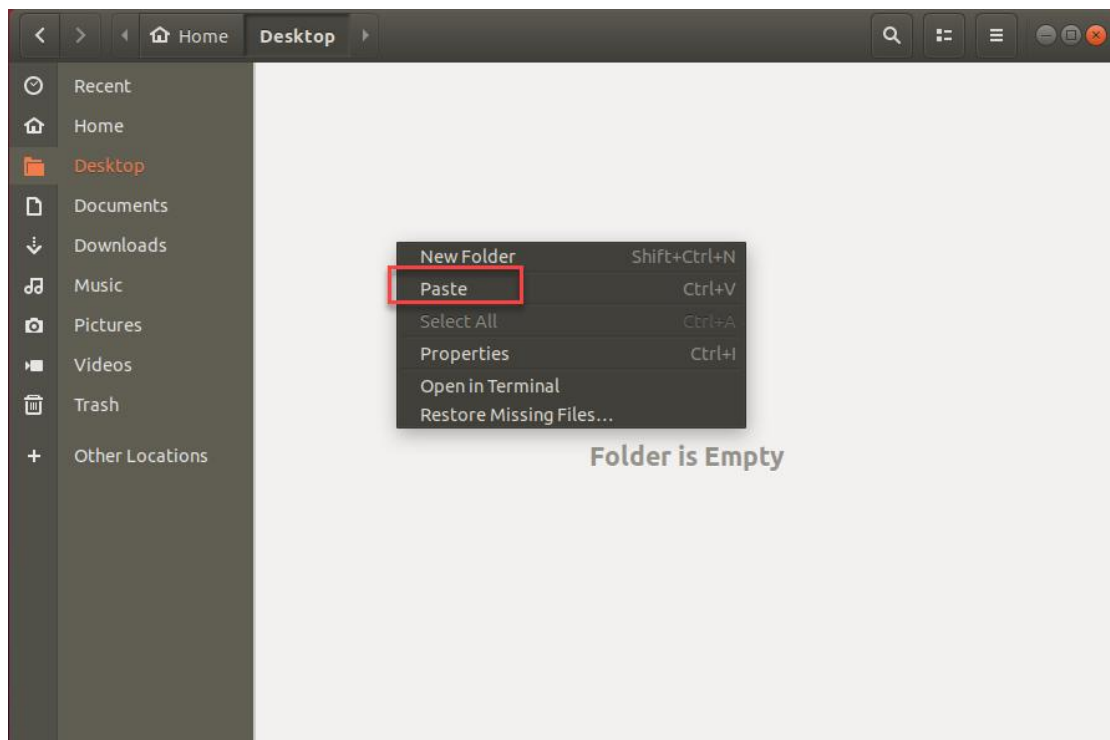
15) Right click **“test”** and click **“copy”**.



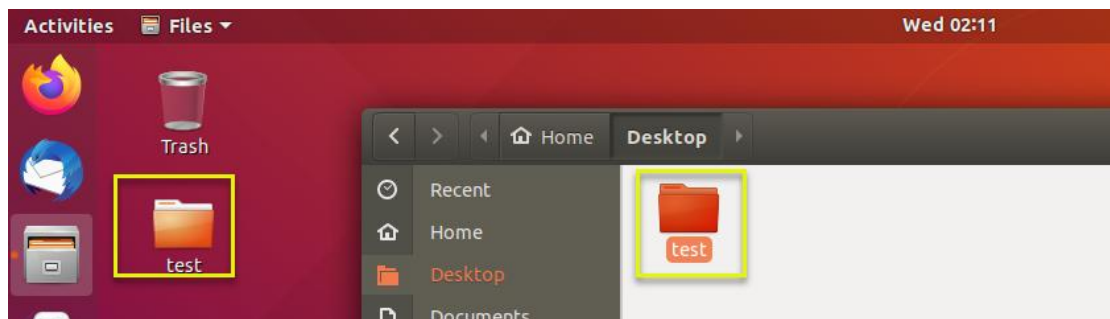
16) Then, click “**desktop**”.



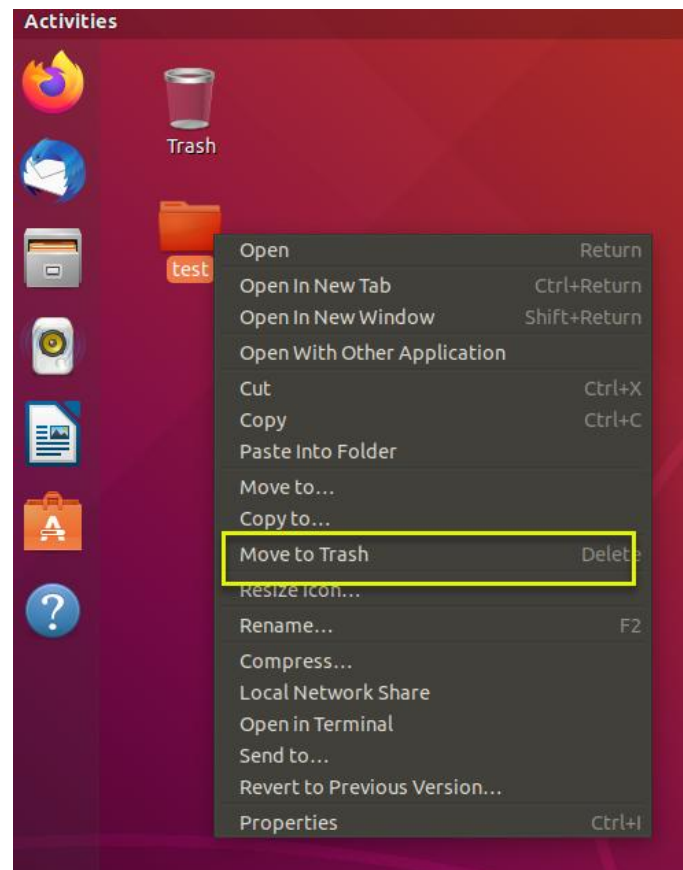
17) Right click the blank area, and then click “**paste**” in the pop-up menu.



18) If “**test**” folder appears on your desktop, it means that the folder is successfully copied.



19) Right click “**test**” folder, and select “**Move to Trash**” to delete the file. And other operation on the folder is the same as that on windows.

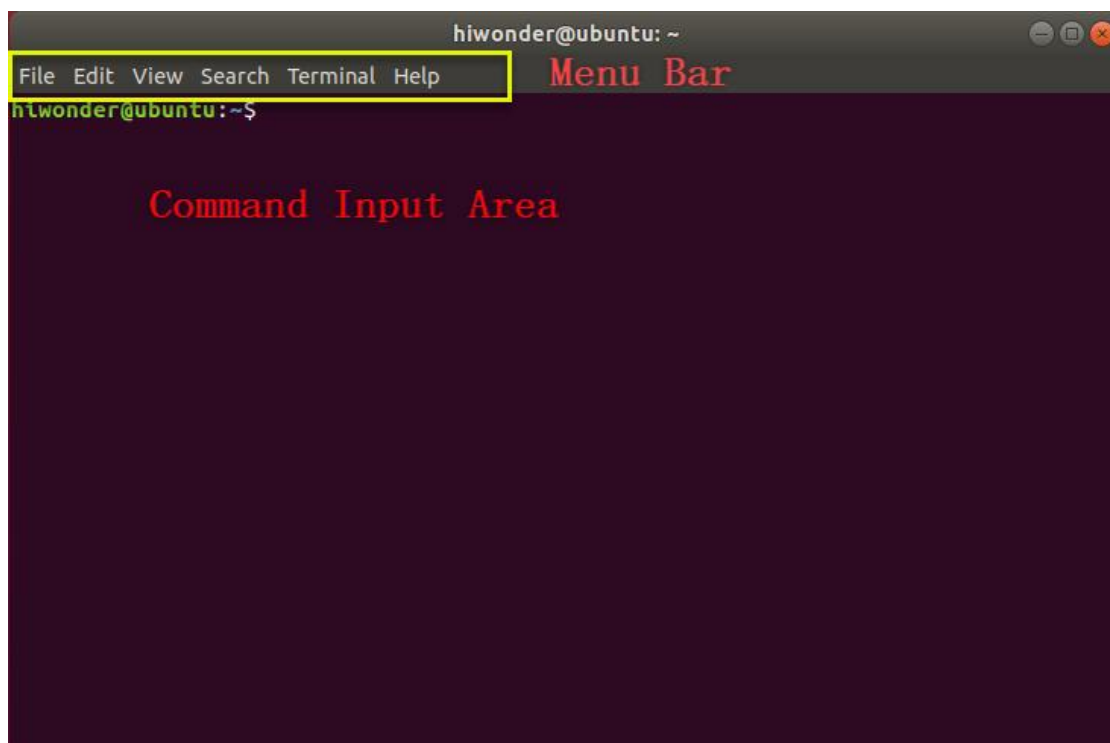


2.Terminal Interface

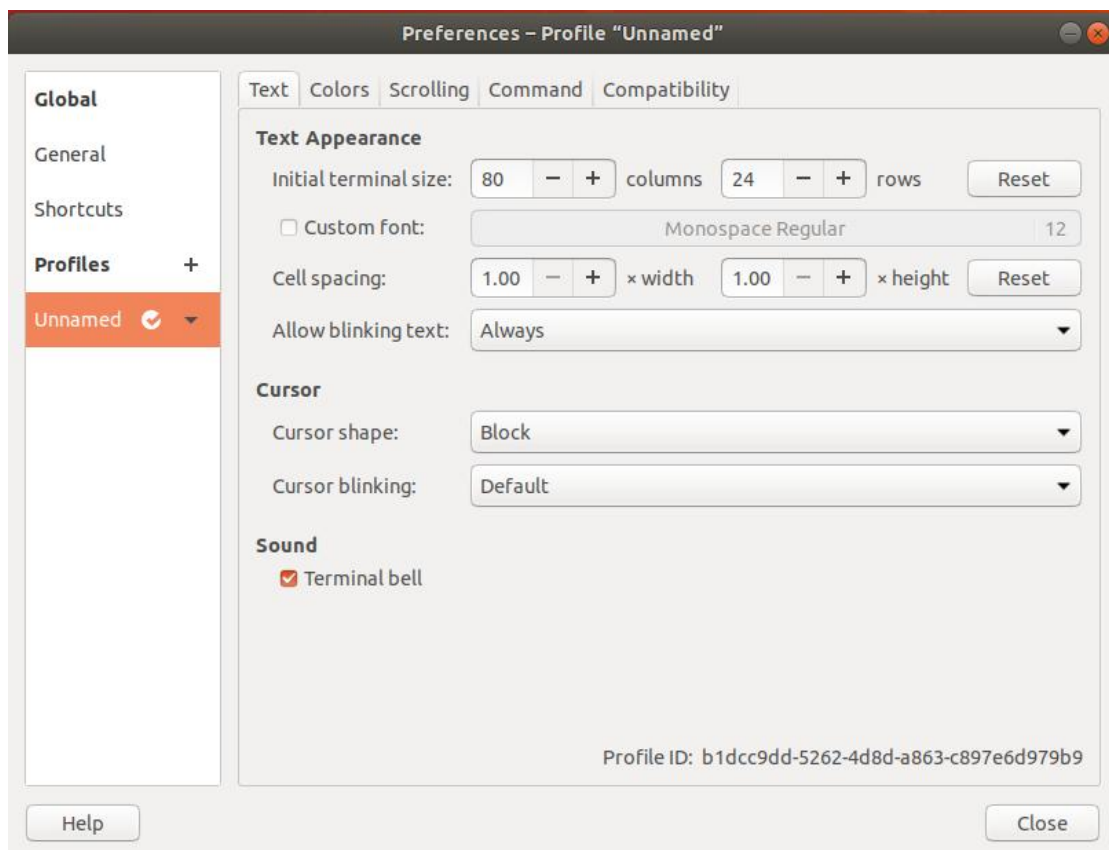
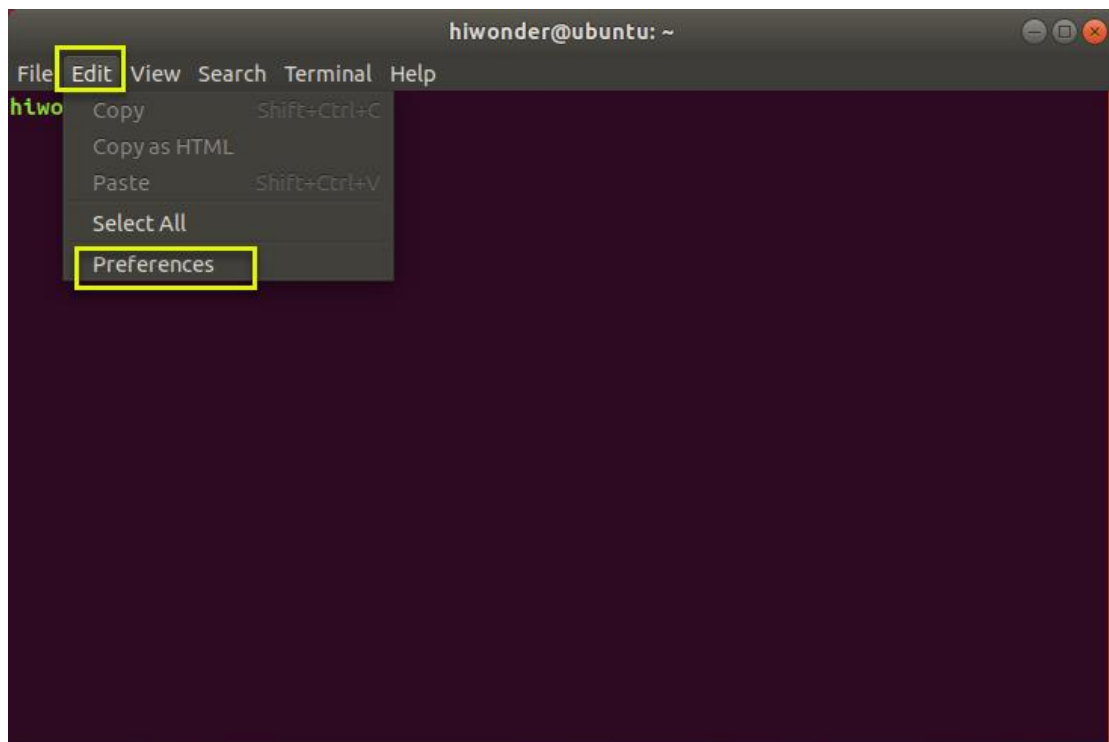
Terminal is the console of Linux, and completing the daily task is relied on the communication between various kinds and the kernel.

The terminal receives input by the user on keyboard and sends the input to the host system, and then the host system processes the user input and returns the result to the terminal.

1) And we can press “**Ctrl+Alt+T**”, or click “” at the bottom left corner and click “**Terminal**” to open the terminal.



2) Click **Edit** and select **Preference**. Then we can change the style, font, etc., at the right side.



3) After entering the terminal interface, you will find the command prompt, such as `[hiwonder@ubuntu~]#`

And hiwonder is the current user name.

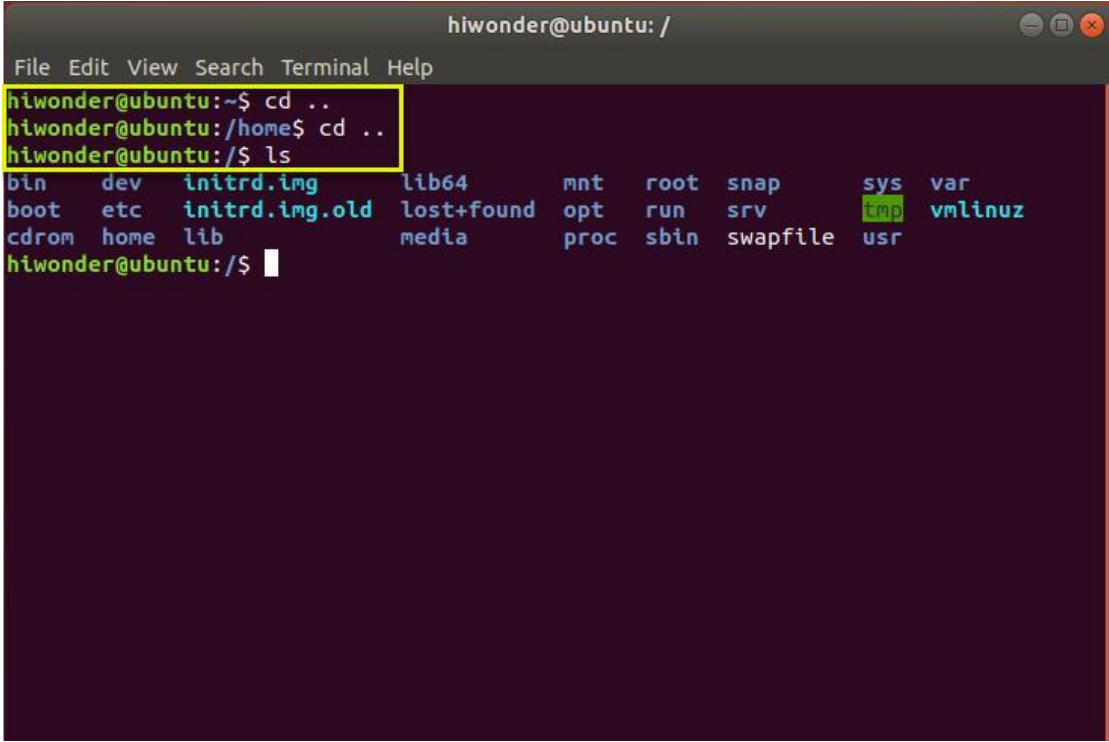
- ① @: separator
- ② Ubuntu: Current host name
- ③ ~: The name of directory the user is currently in. “~” represents home directory.
- ④ #: user identity prompt. # represents super client, that is administrator.
\$ represents normal user.

Lesson 5 Linux Directory Introduction

System directory refers to the directory where the major files of the operating system are stored. The files in the directory has direct influence on the normal function of the system. Therefore, having a basic understanding of these directories will facilitate your usage.

1. Open System Directory

Input “**cd ..**” twice and “**ls**” once in sequence in virtual machine. Note: there is space between “**cd**” and “**..**” , and press **Enter** whenever you finishing inputting.



```

hiwonder@ubuntu: /
File Edit View Search Terminal Help
hiwonder@ubuntu:~$ cd ..
hiwonder@ubuntu:/home$ cd ..
hiwonder@ubuntu:/$ ls
bin      dev      initrd.img  lib64      mnt      root     snap      sys      var
boot     etc      initrd.img.old  lost+found  opt      run      srv        tmp       vmlinuz
cdrom    home     lib         media      proc     sbin     swapfile   usr
hiwonder@ubuntu:/$

```

System directory of Linux refers to the following folders in red frame. Linux’s operating system is completely built upon files and file system. Any information is stored in the form of file and defined by file name and storing path.

Linux’s directory is in the form of dendrogram, and “/” represents the root,

which is also considered as root directory.

```
hiwonder@hiwonder-virtual-machine:/$ ls
bin    dev    initrd.img  lib64    mnt    root  snap    sys    var
boot   etc    initrd.img.old  lost+found  opt    run   srv      tmp    vmlinuz
cdrom  home   lib         media    proc   sbin  swapfile  usr    vmlinuz.old
```

2. Check System Directory

We can check the system directory on Linux through command. For clear demonstration and better understanding, check the system directory in the form of dendrogram.

Input “**sudo apt-get install tree**” command and install the software package.

```
hiwonder@ubuntu:~$ sudo apt-get update
Hit:1 http://mirrors.aliyun.com/ubuntu bionic InRelease
Hit:2 http://mirrors.aliyun.com/ubuntu bionic-updates InRelease
Hit:3 http://mirrors.aliyun.com/ubuntu bionic-backports InRelease
Hit:4 http://mirrors.aliyun.com/ubuntu bionic-security InRelease
Get:5 http://mirrors.aliyun.com/ubuntu bionic/universe i386 Packages [8,531 kB]
Get:6 http://mirrors.aliyun.com/ubuntu bionic/universe amd64 Packages [8,570 kB]
Get:7 http://mirrors.aliyun.com/ubuntu bionic/universe Translation-en [4,941 kB]
Get:8 http://mirrors.aliyun.com/ubuntu bionic/universe amd64 DEP-11 Metadata [3,287 kB]
Get:9 http://mirrors.aliyun.com/ubuntu bionic/universe DEP-11 48x48 Icons [2,151 kB]
Get:10 http://mirrors.aliyun.com/ubuntu bionic/universe DEP-11 64x64 Icons [8,420 kB]
Get:11 http://mirrors.aliyun.com/ubuntu bionic/multiverse amd64 Packages [151 kB]
Get:12 http://mirrors.aliyun.com/ubuntu bionic/multiverse i386 Packages [144 kB]
Get:13 http://mirrors.aliyun.com/ubuntu bionic/multiverse Translation-en [108 kB]
Get:14 http://mirrors.aliyun.com/ubuntu bionic/multiverse amd64 DEP-11 Metadata [49.7 kB]
Get:15 http://mirrors.aliyun.com/ubuntu bionic/multiverse DEP-11 48x48 Icons [8,931 B]
```

Note: if the message below is threw, the installation ends in failure.

```
hiwonder@ubuntu:~$ sudo apt-get install tree
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package tree
```

Input command "**sudo apt-get update**" to update the source.

```
hiwonder@ubuntu:~$ sudo apt-get update
Hit:1 http://mirrors.aliyun.com/ubuntu bionic InRelease
Hit:2 http://mirrors.aliyun.com/ubuntu bionic-updates InRelease
Hit:3 http://mirrors.aliyun.com/ubuntu bionic-backports InRelease
Hit:4 http://mirrors.aliyun.com/ubuntu bionic-security InRelease
Get:5 http://mirrors.aliyun.com/ubuntu bionic/universe i386 Packages [8,531 kB]
Get:6 http://mirrors.aliyun.com/ubuntu bionic/universe amd64 Packages [8,570 kB]
Get:7 http://mirrors.aliyun.com/ubuntu bionic/universe Translation-en [4,941 kB]
Get:8 http://mirrors.aliyun.com/ubuntu bionic/universe amd64 DEP-11 Metadata [3,287 kB]
Get:9 http://mirrors.aliyun.com/ubuntu bionic/universe DEP-11 48x48 Icons [2,151 kB]
Get:10 http://mirrors.aliyun.com/ubuntu bionic/universe DEP-11 64x64 Icons [8,420 kB]
Get:11 http://mirrors.aliyun.com/ubuntu bionic/multiverse amd64 Packages [151 kB]
Get:12 http://mirrors.aliyun.com/ubuntu bionic/multiverse i386 Packages [144 kB]
Get:13 http://mirrors.aliyun.com/ubuntu bionic/multiverse Translation-en [108 kB]
Get:14 http://mirrors.aliyun.com/ubuntu bionic/multiverse amd64 DEP-11 Metadata [49.7 kB]
Get:15 http://mirrors.aliyun.com/ubuntu bionic/multiverse DEP-11 48x48 Icons [8,931 B]
```

After update, input command "**sudo apt-get install tree**" to install again.

After installation, we can use tree related commands to check the directory.

tree: Display all files in the form of dendrogram.

tree -L N: All folders are displayed in the form of dendrogram, and Nth layer of the subfolders will be displayed. (There is a space between "**tree**" and "-", and between "**L**" and "**N**". N needs to be replaced by a specific number which indicates the layer of folder.)

3) Enter the "**tree -L 1**" command to display the subfolders to the first layer, as shown in the figure below. The Windows system also adopts dendrogram, but it takes disk as root partition. The C disk and D disk are equivalent to the first layer of subfolders.

For Linux systems, "/" is equal to integrate disk which is divided into several partitions such as "/etc", "/dev", and "/lib".

```
hiwonder@hiwonder-virtual-machine:/$ tree -L 1
.
├── bin
├── boot
├── cdrom
├── dev
├── etc
├── home
├── initrd.img -> boot/initrd.img-5.4.0-113-generic
├── initrd.img.old -> boot/initrd.img-5.4.0-84-generic
├── lib
├── lib64
├── lost+found
├── media
├── mnt
├── opt
├── proc
├── root
├── run
├── sbin
├── snap
├── srv
├── swapfile
├── sys
├── tmp
├── usr
├── var
├── vmlinuz -> boot/vmlinuz-5.4.0-113-generic
└── vmlinuz.old -> boot/vmlinuz-5.4.0-84-generic

22 directories, 5 files
hiwonder@hiwonder-virtual-machine:/$
```

The function of each directory is listed below.

Directory	Function
bin	Store commonly used Linux commands
boot	Store Linux's startup file.
dev	Store Linux's external device
etc	Store various configuration files and sub directories required by system management
home	Store home directory

<code>lib</code>	Store dynamic link shared libraries.
<code>media</code>	Provide conventional mount points for all removable devices.
<code>mnt</code>	Mount point for temporary files.
<code>proc</code>	Store information about system resources.
<code>root</code>	Home directory of the root user.
<code>sbin</code>	Store non-essential and unimportant system binary files and network application tools in the system.
<code>sys</code>	Store kernel, firmware and system files.
<code>tmp</code>	Store temporary files.
<code>usr</code>	Store user documents, games, graphics files, libraries, other user, management commands and files.
<code>var</code>	Store the frequently modified directory.

Lesson 7 Linux Permission

In operating system, permissions are required to access the file, enter the directory and change the file.

In Linux, different users, including supper user (root) and normal user, have different permission. Super user can do anything on Linux almost without limitation, while normal user can only operate within permissions.

For example, in the same file, super user can edit it, while normal user can only read it without other operations.

1. Change User

It is simple to change user through command. But we need to set a fixed password first, as root password is generated randomly whenever Ubuntu reboots by default.

1) Open command line terminal and input “**sudo passwd**” command. Then input the password and press Enter to set root password. Pay attention you are required to input the password twice. (**Note: the password you input will be hidden and you can just press Enter after inputting**)

```
hiwonder@ubuntu-virtual-machine: ~  
File Edit View Search Terminal Help  
hiwonder@ubuntu-virtual-machine:~$ sudo passwd  
[sudo] password for hiwonder:  
Enter new UNIX password:  
Retype new UNIX password:  
passwd: password updated successfully  
hiwonder@ubuntu-virtual-machine:~$
```

```
hiwonder@ubuntu-virtual-machine:~$ su  
Password:  
root@ubuntu-virtual-machine:/home/hiwonder#
```

2) After setting root password, input “**su**”, and then input the password

currently set and press **Enter**. When \$ at the end of the command changes to #, it means that the directory is switched to root directory.

```
hiwonder@ubuntu-virtual-machine:~$ su
Password:
root@ubuntu-virtual-machine:/home/hiwonder#
```

2. File Permission

Files can be written, read and executed. And files can be divided into three types by visitor:

- ① Owner: user of files
- ② Group: group files belong to
- ③ Other

Owner, group and other indicate roles, such as programmer, while root user and normal user refer to specific person, such as you.

And each role has three permissions.

r: read

read permission grants the ability to read a file. When set for a directory, what is in the directory cannot be viewed.

w: write

The write permission grants the ability to write data to the file. When set for a directory, the files in the directory cannot be deleted.

x: execute

The execute permission grants the ability to execute a file, which is inapplicable to non-executable file. When set for directory, we cannot enter the

directory.

Linux	Octal	Explanation
r--	4	Read permission only
-w-	2	Write permission only
--x	1	Execute permission only
r-x	5	Read and execute permission
rw-	6	Read and write permission
-wx	3	Write and execute permission
rwX	7	Write, read and execute permission

3. Change Permission

chmod command is used to change permission, which is divided into two ways. One is change the permission through number code, the other is change the permission through character code.

1) Open the terminal interface, and input “**touch test.txt**” command to create txt file.

```
hiwonder@ubuntu: ~  
File Edit View Search Terminal Help  
hiwonder@ubuntu:~$ touch test.txt  
hiwonder@ubuntu:~$
```

2) Then enter “**ls -al**” command to list the files under this path and display their specific properties.

```
hiwonder@ubuntu:~$ ls -al
```

```
hiwonder@ubuntu: ~  
File Edit View Search Terminal Help  
-rw-r--r-- 1 hiwonder hiwonder 1276 Jul 6 03:55 .bash_history  
-rw-r--r-- 1 hiwonder hiwonder 220 Jul 5 21:22 .bash_logout  
-rw-r--r-- 1 hiwonder hiwonder 3771 Jul 5 21:22 .bashrc  
drwx----- 15 hiwonder hiwonder 4096 Jul 6 01:54 .cache  
drwx----- 11 hiwonder hiwonder 4096 Jul 6 00:53 .config  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 6 03:02 Desktop  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Documents  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Downloads  
-rw-r--r-- 1 hiwonder hiwonder 8980 Jul 5 21:22 examples.desktop  
drwx----- 3 hiwonder hiwonder 4096 Jul 5 23:37 .gnupg  
-rw-r--r-- 1 hiwonder hiwonder 1916 Jul 6 01:54 .ICEauthority  
drwx----- 3 hiwonder hiwonder 4096 Jul 5 21:40 .local  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Music  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 6 01:56 Pictures  
-rw-r--r-- 1 hiwonder hiwonder 807 Jul 5 21:22 .profile  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Public  
drwx----- 2 hiwonder hiwonder 4096 Jul 5 23:37 .ssh  
-rw-r--r-- 1 hiwonder hiwonder 0 Jul 6 01:08 .sudo_as_admin_successful  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Templates  
drwxrwxr-x 2 hiwonder hiwonder 4096 Jul 6 02:01 test  
-rw-rw-r-- 1 hiwonder hiwonder 0 Jul 6 03:55 test.txt  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 videos  
-rw-rw-r-- 1 hiwonder hiwonder 131 Jul 6 01:32 .xinputrc  
hiwonder@ubuntu:~$
```

As the picture shown above, the owners of the files are granted with read and write permission, while group and other users only have read permission.

For example, we can grant execute permission to group and other users. Input “**chmod 777 test.txt**” command.

```
root@hiwonder-virtual-machine:/home/hiwonder# chmod 777 test.txt  
root@hiwonder-virtual-machine:/home/hiwonder#
```

3) Next, input “**ls -al**” again to check whether the permission is changed.

```

hiwonder@ubuntu: ~
File Edit View Search Terminal Help
-rw----- 1 hiwonder hiwonder 1276 Jul 6 03:55 .bash_history
-rw-r--r-- 1 hiwonder hiwonder 220 Jul 5 21:22 .bash_logout
-rw-r--r-- 1 hiwonder hiwonder 3771 Jul 5 21:22 .bashrc
drwx----- 15 hiwonder hiwonder 4096 Jul 6 01:54 .cache
drwx----- 11 hiwonder hiwonder 4096 Jul 6 00:53 .config
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 6 03:02 Desktop
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Documents
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Downloads
-rw-r--r-- 1 hiwonder hiwonder 8980 Jul 5 21:22 examples.desktop
drwx----- 3 hiwonder hiwonder 4096 Jul 5 23:37 .gnupg
-rw----- 1 hiwonder hiwonder 1916 Jul 6 01:54 .ICEauthority
drwx----- 3 hiwonder hiwonder 4096 Jul 5 21:40 .local
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Music
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 6 01:56 Pictures
-rw-r--r-- 1 hiwonder hiwonder 807 Jul 5 21:22 .profile
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Public
drwx----- 2 hiwonder hiwonder 4096 Jul 5 23:37 .ssh
-rw-r--r-- 1 hiwonder hiwonder 0 Jul 6 01:08 .sudo_as_admin_successful
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Templates
drwxrwxr-x 2 hiwonder hiwonder 4096 Jul 6 02:01 test
-rwxrwxrwx 1 hiwonder hiwonder 0 Jul 6 03:55 test.txt
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 videos
-rw-rw-r-- 1 hiwonder hiwonder 131 Jul 6 01:32 .xinputrc
hiwonder@ubuntu:~$

```

All permissions of these three roles can be changed through number code for one time. For changing permission through character code, please check the operations below.

Compared with number code, it is more flexible to change permission through character code. We can set individual permission for one role, for example enter command “**touch test_1.txt**” to create a txt file named **test_1**

```

hiwonder@ubuntu: ~
File Edit View Search Terminal Help
hiwonder@ubuntu:~$ touch test_1.txt
hiwonder@ubuntu:~$

```

4) Next, input “**ls -al**” command to view the file permission.

```

hiwonder@ubuntu: ~
File Edit View Search Terminal Help
-rw-r--r-- 1 hiwonder hiwonder 220 Jul 5 21:22 .bash_logout
-rw-r--r-- 1 hiwonder hiwonder 3771 Jul 5 21:22 .bashrc
drwx----- 15 hiwonder hiwonder 4096 Jul 6 01:54 .cache
drwx----- 11 hiwonder hiwonder 4096 Jul 6 00:53 .config
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 6 03:02 Desktop
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Documents
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Downloads
-rw-r--r-- 1 hiwonder hiwonder 8980 Jul 5 21:22 examples.desktop
drwx----- 3 hiwonder hiwonder 4096 Jul 5 23:37 .gnupg
-rw----- 1 hiwonder hiwonder 1916 Jul 6 01:54 .ICEauthority
drwx----- 3 hiwonder hiwonder 4096 Jul 5 21:40 .local
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Music
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 6 01:56 Pictures
-rw-r--r-- 1 hiwonder hiwonder 807 Jul 5 21:22 .profile
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Public
drwx----- 2 hiwonder hiwonder 4096 Jul 5 23:37 .ssh
-rw-r--r-- 1 hiwonder hiwonder 0 Jul 6 01:08 .sudo_as_admin_successful
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Templates
drwxrwxr-x 2 hiwonder hiwonder 4096 Jul 6 02:01 test
-rw-rw-r-- 1 hiwonder hiwonder 0 Jul 6 04:00 test_1.txt
-rwxrwxrwx 1 hiwonder hiwonder 0 Jul 6 03:55 test.txt
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Videos
-rw-rw-r-- 1 hiwonder hiwonder 131 Jul 6 01:32 .xinputrc
hiwonder@ubuntu:~$

```

5) Take granting execute permission to owner, group and other users separately for example.

The commands are as follow.

```

chmod u+x test_1.txt

chmod g+x test_1.txt

chmod o+x test_1.txt

```

6) As the same, we can modify x(execute) as r (read) or w(write) to set read or write permission.

```

hiwonder@ubuntu:~$ chmod u+x test_1.txt
hiwonder@ubuntu:~$ chmod g+x test_1.txt
hiwonder@ubuntu:~$ chmod o+x test_1.txt
hiwonder@ubuntu:~$

```

7) After setting, input “ls -al” command to check the properties.

```

hiwonder@ubuntu: ~
File Edit View Search Terminal Help
-rw-r--r-- 1 hiwonder hiwonder 220 Jul 5 21:22 .bash_logout
-rw-r--r-- 1 hiwonder hiwonder 3771 Jul 5 21:22 .bashrc
drwx----- 15 hiwonder hiwonder 4096 Jul 6 01:54 .cache
drwx----- 11 hiwonder hiwonder 4096 Jul 6 00:53 .config
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 6 03:02 Desktop
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Documents
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Downloads
-rw-r--r-- 1 hiwonder hiwonder 8980 Jul 5 21:22 examples.desktop
drwx----- 3 hiwonder hiwonder 4096 Jul 5 23:37 .gnupg
-rw----- 1 hiwonder hiwonder 1916 Jul 6 01:54 .ICEauthority
drwx----- 3 hiwonder hiwonder 4096 Jul 5 21:40 .local
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Music
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 6 01:56 Pictures
-rw-r--r-- 1 hiwonder hiwonder 807 Jul 5 21:22 .profile
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Public
drwx----- 2 hiwonder hiwonder 4096 Jul 5 23:37 .ssh
-rw-r--r-- 1 hiwonder hiwonder 0 Jul 6 01:08 .sudo_as_admin_successful
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Templates
drwxrwxr-x 2 hiwonder hiwonder 4096 Jul 6 02:01 test
-rwxrwxr-x 1 hiwonder hiwonder 0 Jul 6 04:00 test_1.txt
-rwxrwxrwx 1 hiwonder hiwonder 0 Jul 6 03:55 test.txt
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Videos
-rw-rw-r-- 1 hiwonder hiwonder 131 Jul 6 01:32 .xinputrc
hiwonder@ubuntu:~$

```

8) If you don't want to set separately, for example, we can input “**chmod a+w test_1.txt**” command to set write permission for three roles. And “a” represents three roles.

```

hiwonder@ubuntu:~$ chmod a+w test_1.txt
hiwonder@ubuntu:~$

```

9) Next, input “**ls -al**” command to check the properties.

```

hiwonder@ubuntu: ~
File Edit View Search Terminal Help
-rw-r--r-- 1 hiwonder hiwonder 220 Jul 5 21:22 .bash_logout
-rw-r--r-- 1 hiwonder hiwonder 3771 Jul 5 21:22 .bashrc
drwx----- 15 hiwonder hiwonder 4096 Jul 6 01:54 .cache
drwx----- 11 hiwonder hiwonder 4096 Jul 6 00:53 .config
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 6 03:02 Desktop
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Documents
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Downloads
-rw-r--r-- 1 hiwonder hiwonder 8980 Jul 5 21:22 examples.desktop
drwx----- 3 hiwonder hiwonder 4096 Jul 5 23:37 .gnupg
-rw----- 1 hiwonder hiwonder 1916 Jul 6 01:54 .ICEauthority
drwx----- 3 hiwonder hiwonder 4096 Jul 5 21:40 .local
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Music
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 6 01:56 Pictures
-rw-r--r-- 1 hiwonder hiwonder 807 Jul 5 21:22 .profile
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Public
drwx----- 2 hiwonder hiwonder 4096 Jul 5 23:37 .ssh
-rw-r--r-- 1 hiwonder hiwonder 0 Jul 6 01:08 .sudo_as_admin_successful
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Templates
drwxrwxr-x 2 hiwonder hiwonder 4096 Jul 6 02:01 test
-rwxrwxrwx 1 hiwonder hiwonder 0 Jul 6 04:00 test_1.txt
-rwxrwxrwx 1 hiwonder hiwonder 0 Jul 6 03:55 test.txt
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Videos
-rw-rw-r-- 1 hiwonder hiwonder 131 Jul 6 01:32 .xinputrc
hiwonder@ubuntu:~$

```

If want to remove the permission from visitor, we can change “+” as “-”.
For example, remove the execute permission of other users.

10) Input “**chmod o-x test_1.txt**” command.

```

hiwonder@ubuntu:~$ chmod o-x test_1.txt
hiwonder@ubuntu:~$

```

11) Then enter “ls -al” command to view the properties.

```
hiwonder@ubuntu: ~  
File Edit View Search Terminal Help  
-rw-r--r-- 1 hiwonder hiwonder 220 Jul 5 21:22 .bash_logout  
-rw-r--r-- 1 hiwonder hiwonder 3771 Jul 5 21:22 .bashrc  
drwx----- 15 hiwonder hiwonder 4096 Jul 6 01:54 .cache  
drwx----- 11 hiwonder hiwonder 4096 Jul 6 00:53 .config  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 6 03:02 Desktop  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Documents  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Downloads  
-rw-r--r-- 1 hiwonder hiwonder 8980 Jul 5 21:22 examples.desktop  
drwx----- 3 hiwonder hiwonder 4096 Jul 5 23:37 .gnupg  
-rw----- 1 hiwonder hiwonder 1916 Jul 6 01:54 .ICEauthority  
drwx----- 3 hiwonder hiwonder 4096 Jul 5 21:40 .local  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Music  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 6 01:56 Pictures  
-rw-r--r-- 1 hiwonder hiwonder 807 Jul 5 21:22 .profile  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Public  
drwx----- 2 hiwonder hiwonder 4096 Jul 5 23:37 .ssh  
-rw-r--r-- 1 hiwonder hiwonder 0 Jul 6 01:08 .sudo_as_admin_successful  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Templates  
drwxrwxr-x 2 hiwonder hiwonder 4096 Jul 6 02:01 test  
-rw-rwxrwx- 1 hiwonder hiwonder 0 Jul 6 04:00 test 1.txt  
-rw-rwxrwx 1 hiwonder hiwonder 0 Jul 6 03:55 test.txt  
drwxr-xr-x 2 hiwonder hiwonder 4096 Jul 5 21:40 Videos  
-rw-rw-r-- 1 hiwonder hiwonder 131 Jul 6 01:32 .xinputrc  
hiwonder@ubuntu:~$
```